2025

<u>KUKA ROBOT ARM</u>

Teem members:

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4- Moataz Hesham
5- Abdullah Rabie
6- Ahmed Magdy

❖ Robot arm:

❖ Motors:

A1:

A2:

A3:

A4:

A5:

A6:

❖ Drivers:

A1:

A2:

A3:

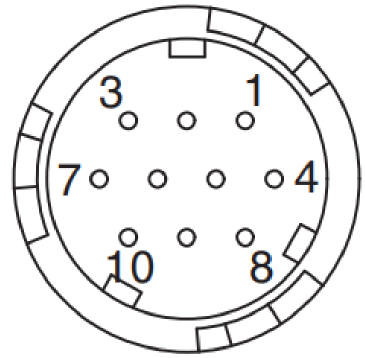
A4:

A5:

A6:

➢ Motors wiring:

1-Cable Specifications for Encoder end Connector (20-bit Encoder):



- Receptacle: CM10-R10P-D.
- Plug: CM10-AP10S-¡-D (Angle)
  CM10SP10S-¡-D (Straight).
- Manufacturer: DDK Ltd.

▪ With an Absolute Encoder:

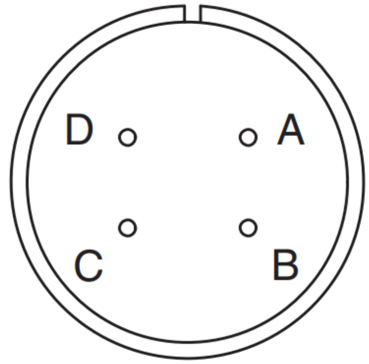
1	PS	6	BAT (+)
2	/PS	7	–
3	–	8	–
4	PG 5V	9	PG 0V
5	BAT (–)	10	FG (Frame ground)

▪ With an Incremental Encoder:

1	PS	6	–
2	/PS	7	–
3	–	8	–
4	PG 5V	9	PG 0V
5	–	10	FG (Frame ground)

2- **Cable Specifications for Servomotor-end Connector:**

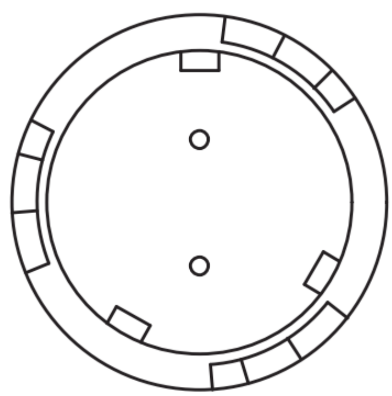
A	Phase U
B	Phase V
C	Phase W
D	FG (Frame ground)



3- **Cable Specifications for Brake-end Connector:**

- Receptacle: CM10-R2P-D.
- Plug: CM10-AP2S-¡-D (Angle)
  CM10-SP2S-¡-D (Straight).
- Manufacturer: DDK Ltd.

Brake terminal
Brake terminal

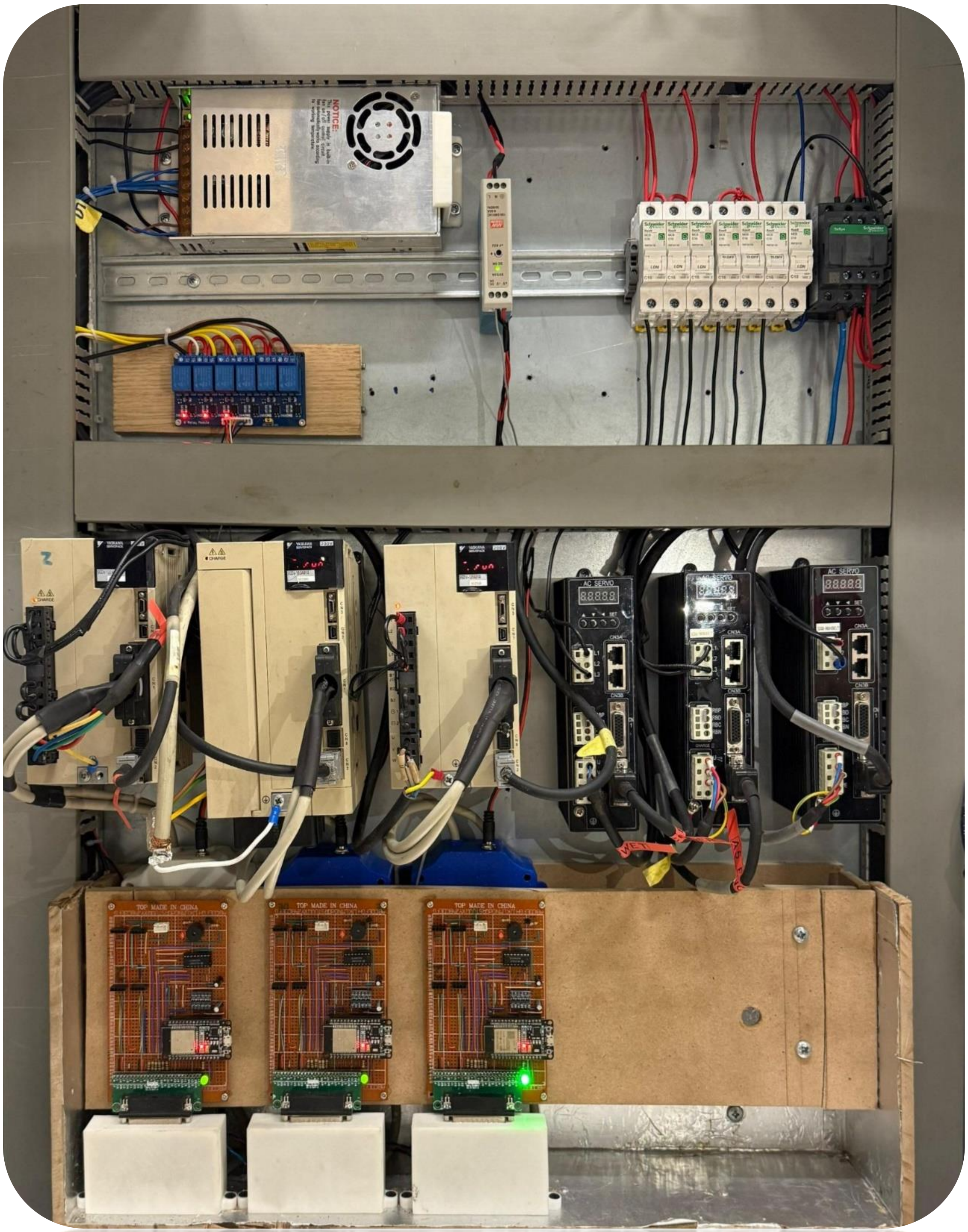


Note: No polarity for connection to the brake terminals.

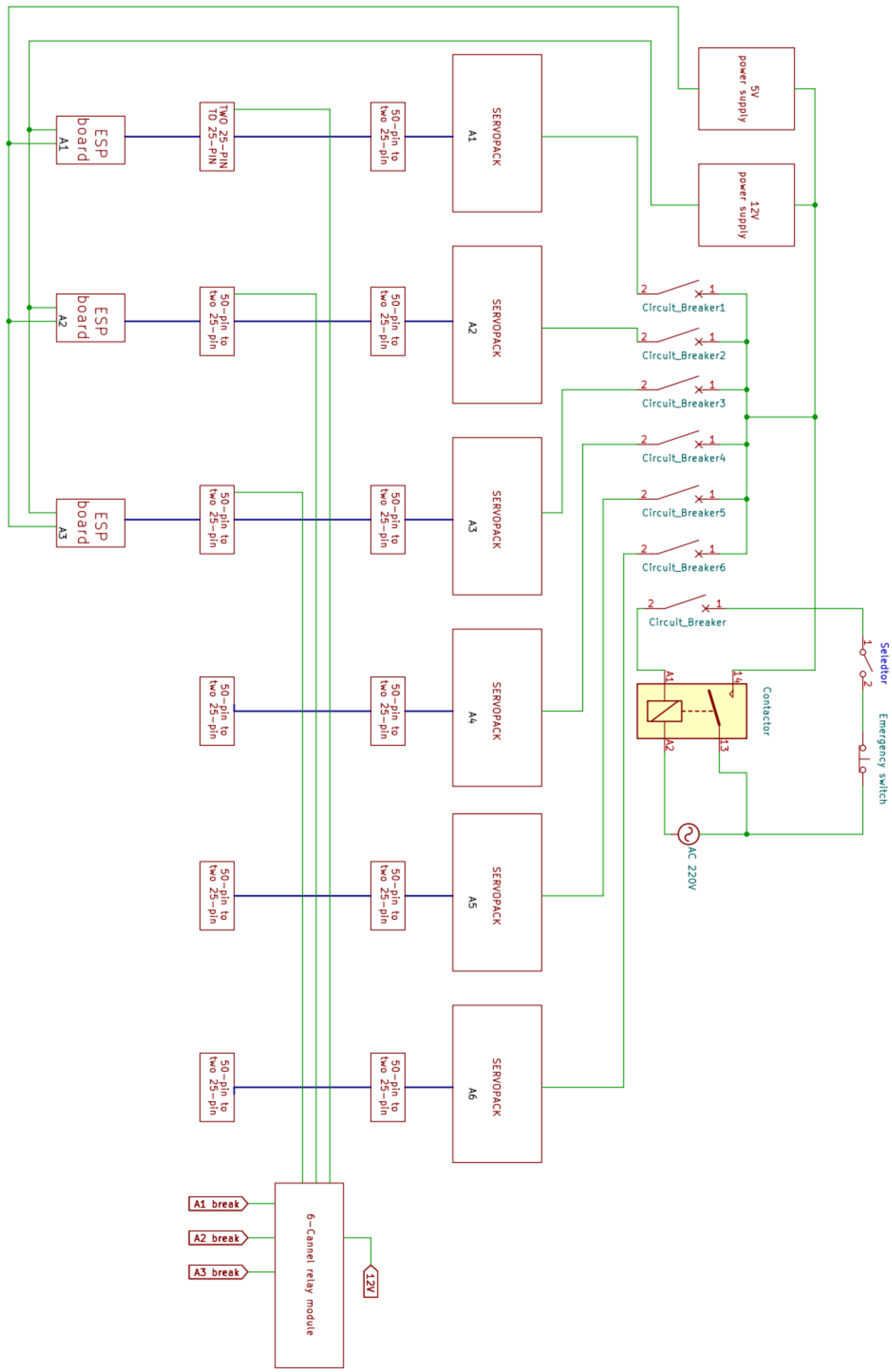
Breaks work on 24V DC.

Make sure all shields are connected.

➢ Control panel:



- **Panel schematic:**



Driver I/O Signal:

1-Input signals:

<table>
<tr><th>Control Method</th><th>Signal Name</th><th>Pin No.</th><th colspan="2">Function</th></tr>
<tr><td rowspan="14">Common</td><td>/S-ON</td><td>40</td><td colspan="2">Servo ON/OFF: Turns ON/OFF the servomotor.</td></tr>
<tr><td rowspan="5">/P-CON</td><td rowspan="5">41</td><td>Proportional control reference</td><td>Switches the speed control loop from PI (proportional/integral) to P (proportional) control when ON.</td></tr>
<tr><td>Rotation Direction reference</td><td>With internal set speed control selected: Switches the servomotor rotation direction.</td></tr>
<tr><td>Control switching</td><td>Position ↔ speed
Position ↔ torque
Torque ↔ speed
Enables control switching.</td></tr>
<tr><td>Zero-clamp reference</td><td>With speed control with zero-clamp function selected:
Reference speed is zero when ON.</td></tr>
<tr><td>Reference pulse block</td><td>With position control with reference pulse stop selected: Stops reference pulse input when ON.</td></tr>
<tr><td>P-OT
N-OT</td><td>42
43</td><td>Forward run prohibited, Reverse run prohibited</td><td>With overtravel prevention: Stops servomotor when movable part travels beyond the allowable range of motion.</td></tr>
<tr><td rowspan="2">/P-CL
/N-CL</td><td rowspan="2">45
46</td><td>Forward external torque limit, Reverse external torque limit</td><td>Activates/deactivates external torque limit function.</td></tr>
<tr><td>Internal set speed switching</td><td>With internal set speed control selected: Switches the internal set speed settings.</td></tr>
<tr><td>/ALM-RST</td><td>44</td><td colspan="2">Alarm reset: Releases the servo alarm state.</td></tr>
<tr><td>+24VIN</td><td>47</td><td colspan="2">Control power supply input for sequence signals.
Allowable voltage range: 11 to 25 V
Note: The 24 VDC power supply is not included.</td></tr>
<tr><td>SEN</td><td>4 (2)</td><td colspan="2">Initial data request signal when using an absolute encoder.</td></tr>
<tr><td>BAT (+)
BAT (-)</td><td>21
22</td><td colspan="2">Connecting pin for the absolute encoder backup battery.
Do not connect when the encoder cable with the battery case is used.</td></tr>
<tr><td>/SPD-D
/SPD-A
/SPD-B
/C-SEL
/ZCLAMP
/INHIBIT
/G-SEL
/PSEL</td><td>Signals that can be allocated</td><td colspan="2">The following input signals can be changed to allocate functions:
/S-ON, /P-CON, P-OT, N-OT, /P-CL, /N-CL, and /ALM-RST.</td></tr>
<tr><td>Speed</td><td>V-REF</td><td>5 (6)</td><td colspan="2">Inputs speed reference. Input voltage range: ± 12 V max.</td></tr>
</table>

Control Method	Signal Name	Pin No.	Function
Position	PULS /PULS SIGN /SIGN	7 8 11 12	Input pulse modes: Select one of them. • Sign + pulse train • CW + CCW pulse train • Two-phase pulse train with 90° phase differential
	CLR /CLR	15 14	Clears position error during position control.
Torque	T-REF	9 (10)	Inputs torque reference. Input voltage range: ± 12 V max.

Note: Pin numbers in parentheses () indicate signal grounds.

2-Output signals:

<table>
<tr><th>Control Method</th><th>Signal Name</th><th>Pin No.</th><th colspan="2">Function</th></tr>
<tr><td rowspan="10">Common</td><td>ALM+
ALM-</td><td>31
32</td><td colspan="2">Servo alarm: Turns OFF when an error is detected.</td></tr>
<tr><td>/TGON+
/TGON-</td><td>27
28</td><td colspan="2">Detection during servomotor rotation: Turns ON when the servomotor is rotating at a speed higher than the motor speed setting.</td></tr>
<tr><td>/S-RDY+
/S-RDY-</td><td>29
30</td><td colspan="2">Servo ready: Turns ON when the SERVOPACK is ready to accept the servo ON (/S-ON) signal.</td></tr>
<tr><td>PAO
/PAO</td><td>33
34</td><td>Phase-A signal</td><td rowspan="2">Encoder output pulse signals with 90° phase differential</td></tr>
<tr><td>PBO
/PBO</td><td>35
36</td><td>Phase-B signal</td></tr>
<tr><td>PCO
/PCO</td><td>19
20</td><td>Phase-C signal</td><td>Origin pulse output signal</td></tr>
<tr><td>ALO1
ALO2
ALO3</td><td>37 (1)
38 (1)
39 (1)</td><td colspan="2">Alarm code output: Outputs 3-bit alarm codes.</td></tr>
<tr><td>FG</td><td>Shell</td><td colspan="2">Connected to frame ground if the shielded wire of the I/O signal cable is connected to the connector shell.</td></tr>
<tr><td>/CLT
/VLT
/BK
/WARN
/NEAR
/PSELA</td><td>Signals that can be allocated</td><td colspan="2">The following output signals can be changed to allocate functions: /TGON, /S-RDY, and /V-CMP (/COIN).</td></tr>
<tr><td>Speed</td><td>/V-CMP+
/V-CMP-</td><td>25
26</td><td colspan="2">If speed control is selected, the signal turns ON when the motor speed is within the setting range and it matches the reference speed value.</td></tr>
</table>

Position	/COIN+ /COIN-	25 26	If position control is selected, the signal turns ON when the number of position error reaches the value set.
	PL1 PL2 PL3	3 13 18	Output signals of power supply for open-collector reference
–	–	16 17 23 24 48 49 50	Do not use these pins.

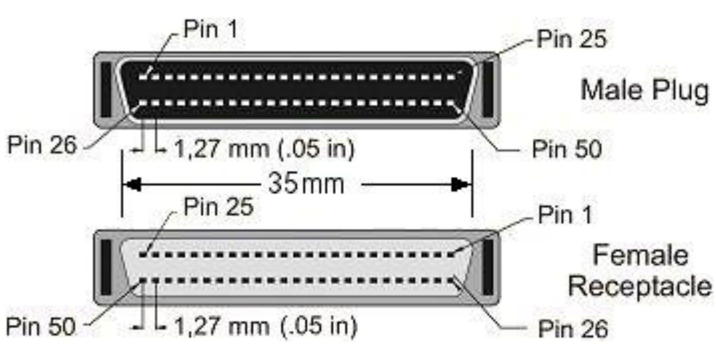
Notes: 1. Pin numbers in parentheses () indicate signal grounds.

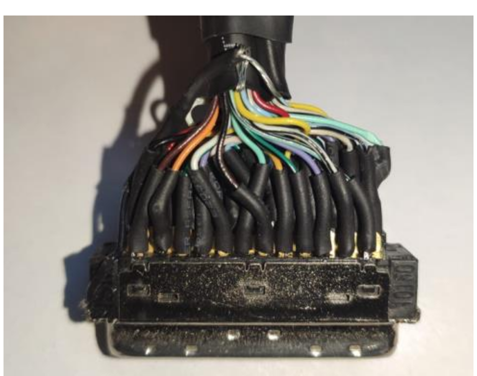
2. The functions allocated to /TGON, /S-RDY, and /V-CMP (/COIN) output signals can be changed by using the parameters. Refer to 3.3.2 Output Signal Allocations for details.

3. We use pin (27,28) to activate external break.

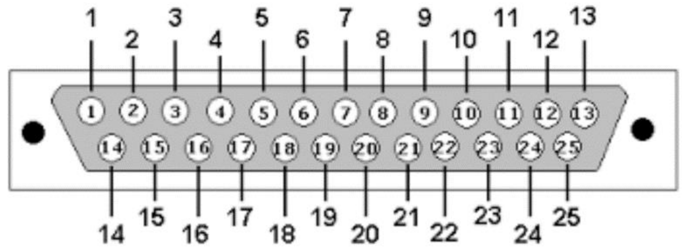
50-pin to two 25-pin:

- YASKAWA original 50-pin connector:

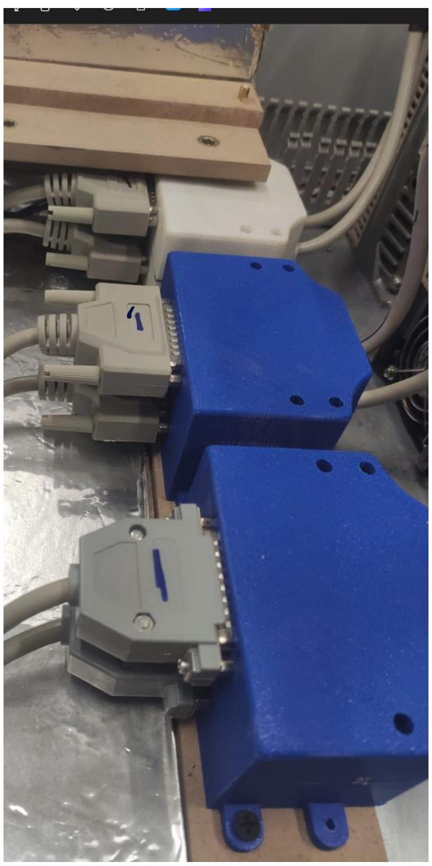


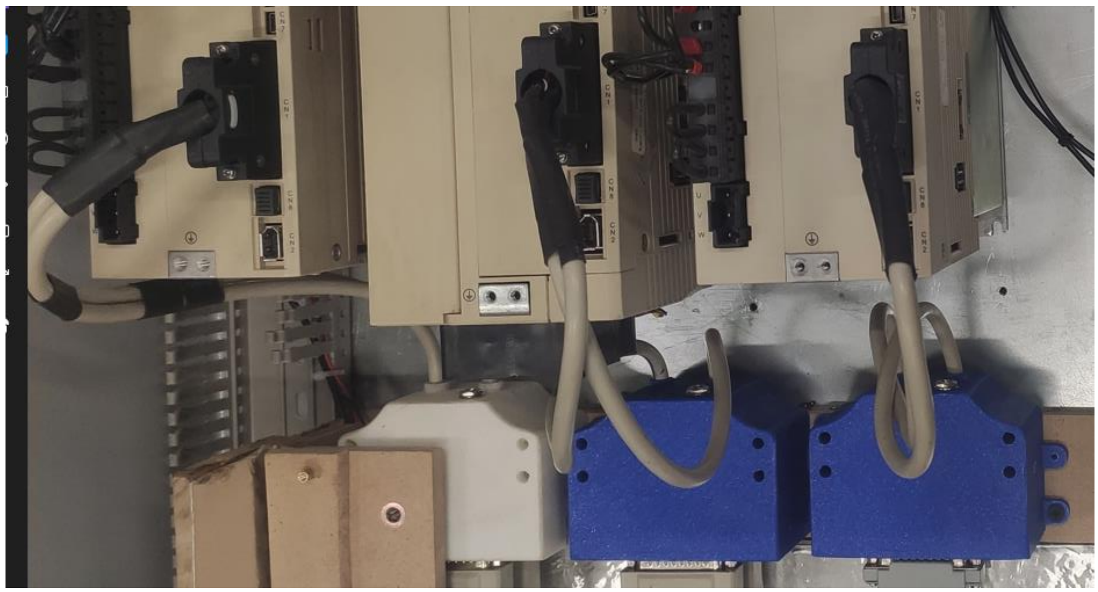


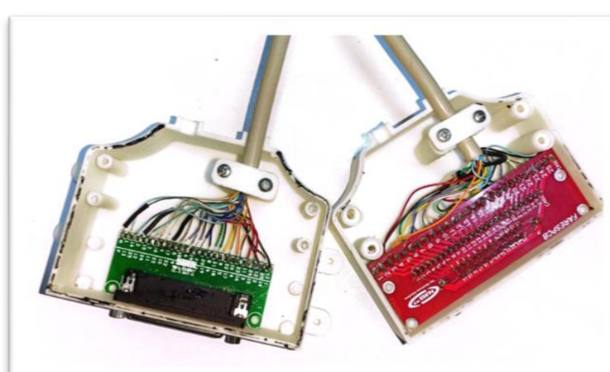
- DB25 25-pin connector:



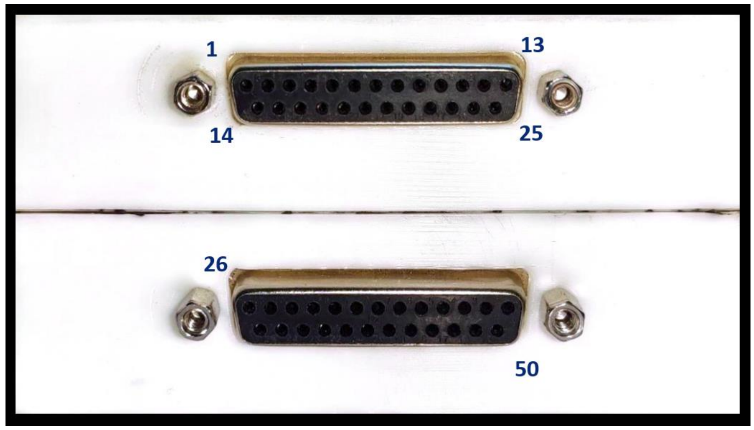
- From 50-pin to two 25-pin:



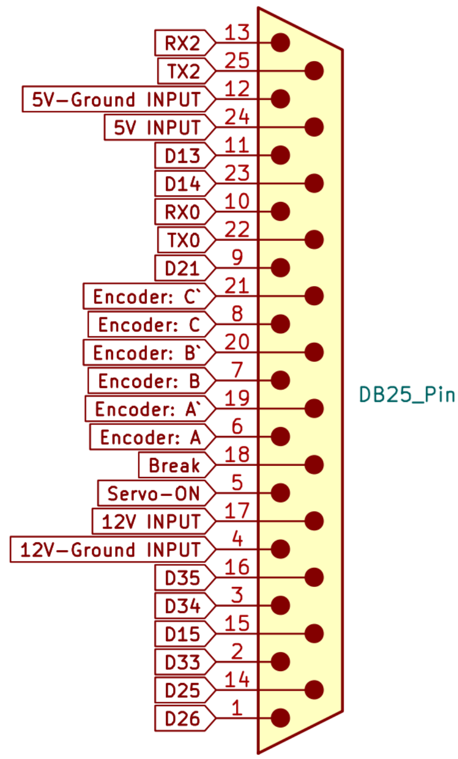




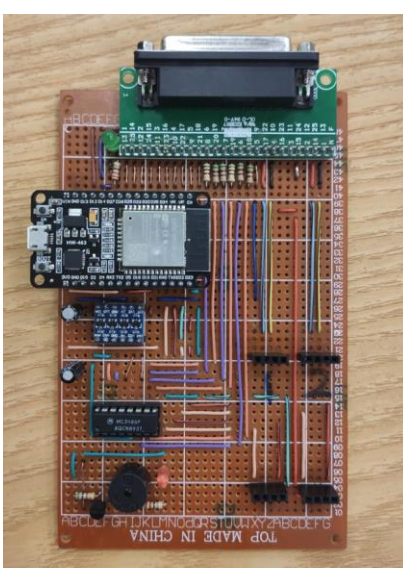
- pins configuration:

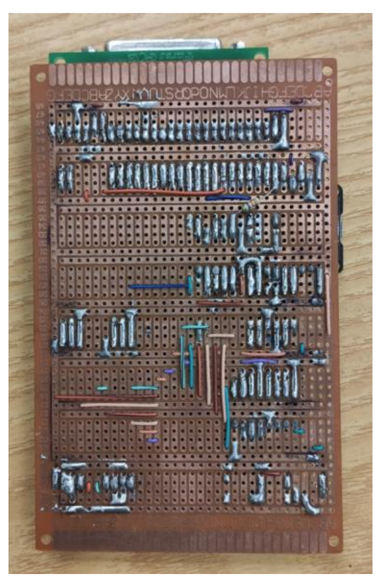


Two 25-pin to 25-pin:



ESP board :

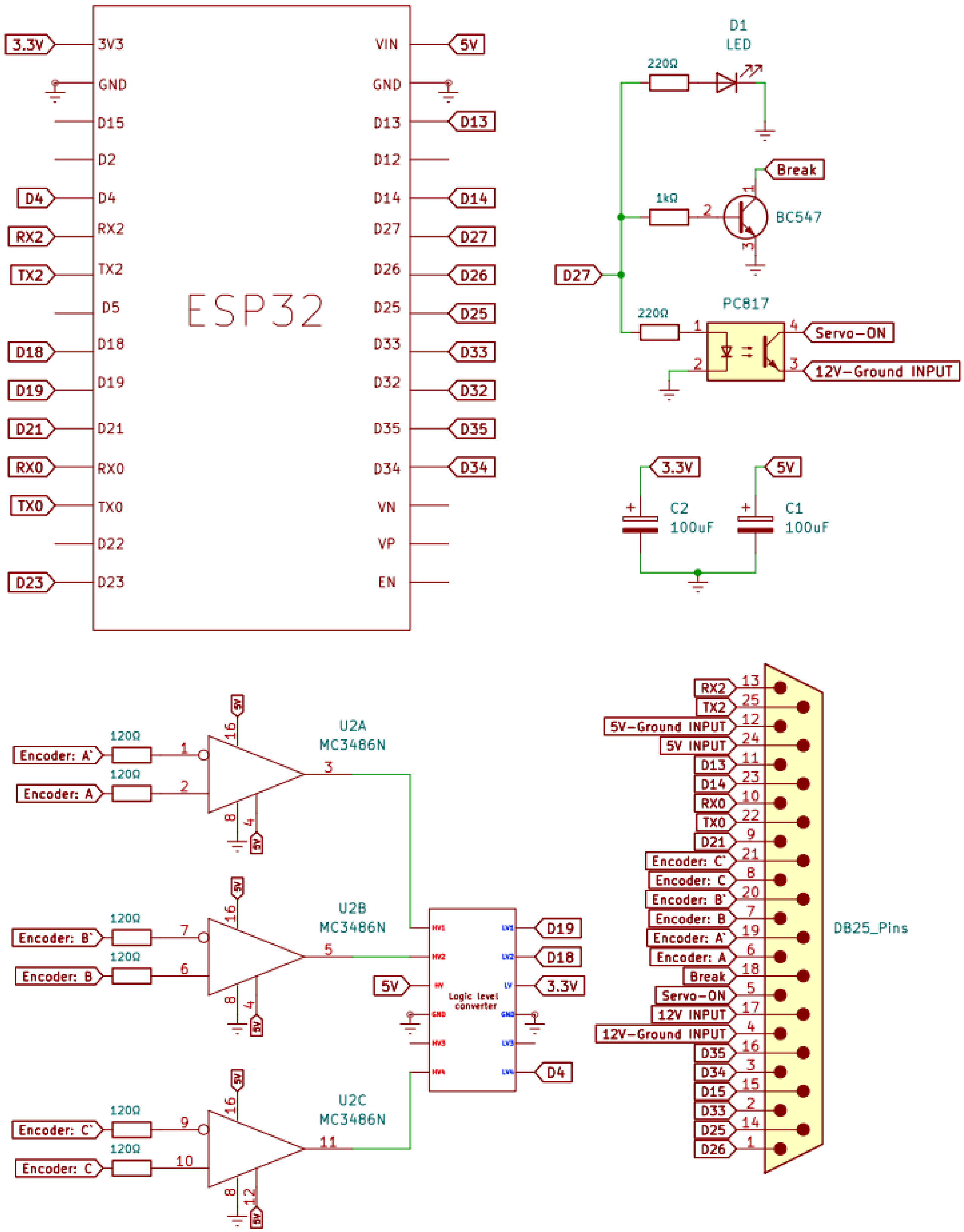




I/O TABLE

ESP	DB25 connector	Function
13	11 (Directly connected to ESP)	Pulses
14	23 (Directly connected to ESP)	Sign
21	9 (Directly connected to ESP)	CLR
25	14 (Directly connected to ESP)	T_{Ref}
26	1 (Directly connected to ESP)	
32	15 (Directly connected to ESP)	
33	2 (Directly connected to ESP)	
34	3	Limit switch
35	16	Limit switch
27	5 (servo-on)	Servo-on, Break-off (active high)
	18 (break)	
19 (A)	6	Encoder: (Non-inverted_A)
	19	Encoder: (Inverted_A)
18 (B)	7	Encoder: (Non-inverted_B)
	20	Encoder: (Inverted_B)
4 (C)	8	Encoder: (Non-inverted_C)
	21	Encoder: (Inverted_C)
23		Buzzer
RX2-TX2	13	RS-485 Transmitter (A)
	25	RS-485 Transmitter (B)
RX0-TX0	10	RS-485 Receiver (A)
	22	RS-485 Receiver (B)
	24	5V INPUT
	12	5V-Ground INPUT
	17	12V INPUT
	4	12V-Ground INPUT
	Shell	5V-Ground

Schematic:



ESP32
3V3
GND
D15
D2
D4
RX2
TX2
D5
D18
D19
D21
RX0
TX0
D22
D23
VIN
D13
D12
D14
D27
D26
D25
D33
D32
D35
D34
VN
VP
EN
3.3V
5V
D1
LED
220Ω
1kΩ
BC547
Break
PC817
Servo−ON
12V−Ground INPUT
C2
100uF
C1
100uF
U2A
MC3486N
U2B
MC3486N
U2C
MC3486N
120Ω
Encoder: A`
Encoder: A
Encoder: B`
Encoder: B
Encoder: C`
Encoder: C
Logic level converter
DB25_Pins
5V−Ground INPUT
5V INPUT
12V INPUT